Mã nguồn chương trình cảm biến HTTP

Bài 1 Phần 1

Hai chương trình Python dưới đây trao đổi dữ liệu cảm biến giả lập qua HTTP GET. Sao chép từng phần thành file có tên tương ứng và giữ nguyên thụt lề.

1 Mã nguồn server

Tên file: server.py

```
"""Server HTTP cung cap du lieu nhiet do va do am gia lap ngau nhien."""

import argparse
import json
import random
from http.server import BaseHTTPRequestHandler, ThreadingHTTPServer
from urllib.parse import urlsplit

TEMPERATURE_RANGE = (20.0, 35.0)  # Don vi: do C.
HUMIDITY_RANGE = (40, 80)  # Don vi: %.


def generate_sensor_data():
    """Sinh du lieu cam bien gia lap moi cho moi request GET."""
    return {
        "temperature": round(random.uniform(*TEMPERATURE_RANGE), 1),
        "humidity": random.randint(*HUMIDITY_RANGE),
    }


class SensorHandler(BaseHTTPRequestHandler):
    def do_GET(self):
        path = urlsplit(self.path).path
        if path == "/api/sensor":
            # Tao du lieu moi, chuyen thanh JSON va gui ve client.
            body = json.dumps(generate_sensor_data()).encode("utf-8")
            content_type = "application/json; charset=utf-8"
        else:
            self.send_error(404, "Endpoint not found")
            return

        self.send_response(200)
        self.send_header("Content-Type", content_type)
        self.send_header("Content-Length", str(len(body)))
        self.send_header("Cache-Control", "no-store")
        self.end_headers()
        self.wfile.write(body)


def main():
    parser = argparse.ArgumentParser(description="Server cam bien HTTP - Bai 1")
    parser.add_argument("--host", default="127.0.0.1")
    parser.add_argument("--port", type=int, default=8000)
    args = parser.parse_args()
    try:
        with ThreadingHTTPServer((args.host, args.port), SensorHandler) as server:
            print(f"Server: http://{args.host}:{args.port}", flush=True)
            print("API: GET /api/sensor -> du lieu gia lap ngau nhien moi", flush=True)
            print("Nhiet do: 20-35 do C; do am: 40-80%.", flush=True)
            print("Nhan Ctrl+C de dung server.", flush=True)
            server.serve_forever()
```

```
    except KeyboardInterrupt:
        print("\nDa dung server.")
    except OSError as exc:
        parser.exit(1, f"Khong khoi dong duoc server: {exc}\n")


if __name__ == "__main__":
    main()
```

2 Mã nguồn client

Tên file: client.py

```
"""Client Python: gui HTTP GET va hien thi du lieu ngau nhien nhan tu server."""

import argparse
import json
import math
from urllib.error import HTTPError, URLError
from urllib.request import Request, urlopen


def read_sensor(url):
    request = Request(url, method="GET", headers={"Accept": "application/json"})
    with urlopen(request, timeout=10) as response:
        status = response.status
        data = json.loads(response.read().decode("utf-8"))
    if not isinstance(data, dict):
        raise ValueError("Du lieu JSON phai la mot object.")
    for field in ("temperature", "humidity"):
        value = data.get(field)
        if isinstance(value, bool) or not isinstance(value, (int, float)):
            raise ValueError(f"Truong {field} phai la so.")
        if not math.isfinite(value):
            raise ValueError(f"Truong {field} phai la so huu han.")
    if not 0 <= data["humidity"] <= 100:
        raise ValueError("Do am phai nam trong khoang 0 den 100%.")
    return status, data


def main():
    parser = argparse.ArgumentParser(description="Client HTTP GET - Bai 1")
    parser.add_argument("--url", default="http://127.0.0.1:8000/api/sensor")
    args = parser.parse_args()
    try:
        print(f"Request: GET {args.url}")
        status, data = read_sensor(args.url)
        print(f"HTTP status: {status} OK")
        print("JSON:", json.dumps(data, ensure_ascii=False))
        print(f"Nhiet do (temperature): {data['temperature']:g} do C")
        print(f"Do am (humidity): {data['humidity']:g}%")
    except HTTPError as exc:
        parser.exit(1, f"Loi HTTP: {exc.code}\n")
    except (URLError, TimeoutError, OSError) as exc:
        parser.exit(1, f"Loi ket noi: {exc}\n")
    except (ValueError, UnicodeError) as exc:
        parser.exit(1, f"Loi du lieu: {exc}\n")


if __name__ == "__main__":
    main()
```